

MATHDNN_1_5

September 9, 2021

```
[1]: import numpy as np

[2]: from IPython.display import set_matplotlib_formats
set_matplotlib_formats('pdf', 'svg')

[3]: class Convolution1d :
    def __init__(self, filt) :
        self.__filt = filt
        self.__r = filt.size
        self.T = TransposedConvolution1d(self.__filt)

    def __matmul__(self, vector) :
        r, n = self.__r, vector.size

        return np.asarray([sum([self.__filt[k]*vector[k+i] for k in range(r)])
        →for i in range(n-r+1)]) # IMPLEMENT THIS

[4]: class TransposedConvolution1d :

    # Transpose of 1-dimensional convolution operator used for the
    →transpose-convolution operation  $A.T@(\dots)$ 

    def __init__(self, filt) :
        self.__filt = filt
        self.__r = filt.size

    # Goal : Implementing matrix multiplication

    def __matmul__(self, vector) :
        r = self.__r
        n = vector.size + r - 1

        return np.asarray([sum([self.__filt[k]*vector[i-k] for k in range(0,i+1) if
        →k<=r-1 and i-k>=0 and i-k<=n-r]) for i in range(n)]) # IMPLEMENT THIS

[5]: def huber_loss(x) :
    return np.sum( (1/2)*(x**2)*(np.abs(x)<=1) + (np.sign(x)*x-1/2)*(np.
    →abs(x)>1) )
```

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def huber_grad(x) :
    return x*(np.abs(x)<=1) + np.sign(x)*(np.abs(x)>1)
```

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[6]: r, n, lam = 3, 20, 0.1

np.random.seed(0)
k = np.random.randn(r) # randn(r) : normalized seeds of r elements
b = np.random.randn(n-r+1)
A = Convolution1d(k)
#from scipy.linalg import circulant
#A = circulant(np.concatenate((np.flip(k),np.zeros(n-r))))[2:,:]
```

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x = np.zeros(n)
alpha = 0.01
for _ in range(100) :
    x = x - alpha*(A.T@(huber_grad(A@x-b))+lam*x)

print(huber_loss(A@x-b)+0.5*lam*np.linalg.norm(x)**2)
```

0.4587586843129764

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[7]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
[ ]: !wget -nc https://raw.githubusercontent.com/brpy/colab-pdf/master/colab_pdf.py
from colab_pdf import colab_pdf
colab_pdf('MATHDNN_1_5.ipynb')
```

```
--2021-09-09 13:16:07-- https://raw.githubusercontent.com/brpy/colab-
pdf/master/colab_pdf.py
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1864 (1.8K) [text/plain]
Saving to: colab_pdf.py
```

```
colab_pdf.py          100%[=====>]    1.82K  --.-KB/s    in 0s
```

```
2021-09-09 13:16:07 (23.0 MB/s) - colab_pdf.py saved [1864/1864]
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